King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 101 Major Exam II Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

---

1. If M and m are the absolute maximum and absolute minimum values of the function $f(x) = 3x^4 - 4x^3$ where $-1 \leq x \leq 2$ respectively, then $M + m =$

(a) 15

(b) 10

(c) 23

(d) 12

(e) 25

2. If a man 1.5 meters tall walks at a rate of 3 meters per second away from a light that is 6 meters above the ground, then the rate at which the length of his shadow is changing when he is 6 meters from the base of the light is equal to

(a) 2 m/sec

(b) 1.5 m/sec

(c) 1 m/sec

(d) 3 m/sec

(e) 2.5 m/sec

3. The x-intercept of the tangent line to the curve of $x^{2/3} + y^{2/3} = 5$ at the point $(1, 8)$ is

(a) $(9, 0)$

(b) $(7, 0)$

(c) $(5, 0)$

(d) $(6, 0)$

(e) $(4, 0)$

4. If the slope of the tangent line to the curve $y = \dfrac{e^x - a}{e^x + a}$ at $x = 0$ is $\dfrac{1}{2}$, then the value of a is equal to

(a) $\dfrac{1}{2}$

(b) -1

(c) 0

(d) $-\dfrac{1}{2}$

(e) 1

5. The slope of the tangent line to the curve $x^4 + 2x^2y^2 + y^4 = 4x^2y$ at $(-1, 1)$ is

(a) 0

(b) -1

(c) -2

(d) 2

(e) 1

6. If a point moves along the curve $y = \sqrt[3]{x^2}$ such that the y-component of the point is increasing at a rate of 2 units per second, then the rate at which the x-component is changing when $x = 1$ is equal to

(a) 1

(b) 3

(c) $\dfrac{2}{3}$

(d) $\dfrac{3}{2}$

(e) 6

7. If f is a function such that $f'(x) \geq 4$ for all $x \in (-2, 2)$ and $f(1) = 3$, then the largest value that $f(0)$ can take is

(a) 3

(b) 4

(c) -1

(d) 0

(e) 2

8. If $f(x) = 2\sin x - \cos 2x$ is defined on $[0, 2\pi]$, then the sum of all its critical numbers is

(a) $\frac{5\pi}{3}$

(b) 3π

(c) $\frac{9\pi}{2}$

(d) 2π

(e) 5π

9. If $f(x) = \frac{(2x+1)^3(x^2-1)^2}{x+3}$, then $f'(0) =$

(a) $\frac{17}{9}$

(b) $\frac{19}{7}$

(c) $\frac{21}{5}$

(d) $\frac{15}{7}$

(e) $\frac{13}{3}$

10. If the normal line to the parabola $y = \frac{1}{2}(x-2)^2 + 1$ at the point $\left(3, \frac{3}{2}\right)$ intersects the parabola at another point (a, b) then $4a + 2b =$

(a) 7

(b) 0

(c) 1

(d) 3

(e) 13

11. The function $f(x) = x^3 + 2x^2 + x + 1$ has

(a) a local maximum at $x = -\frac{2}{3}$ and a local minimum at $x = -\frac{1}{3}$

(b) a local maximum at $x = -1$ and a local minimum at $x = -\frac{1}{3}$

(c) a local maximum at $x = -\frac{4}{3}$ and a local minimum at $x = -\frac{2}{3}$

(d) a local minimum at $x = -\frac{2}{3}$ and a local maximum at $x = -\frac{1}{3}$

(e) a local minimum at $x = -1$ and a local maximum at $x = -\frac{1}{3}$

12. If $f(x) = x^3 + 3x - 1$, then $(f^{-1})'(-5) =$

(a) $\frac{2}{5}$

(b) $\frac{2}{3}$

(c) $\frac{1}{2}$

(d) $\frac{1}{3}$

(e) $\frac{1}{6}$

13. If $y(x) = \ln \sqrt{\frac{1-x}{1+x}} + \arctan x$, then $y'(0) =$

(a) 1

(b) -2

(c) 2

(d) 0

(e) -1

14. If $f(x) = x \log_2 x$ is defined on the closed interval $[1, 2]$, then a number c in the open interval $(1, 2)$ such that $f'(c) = f(2) - f(1)$ is equal to

(a) $\frac{5}{e}$

(b) $\frac{3}{e}$

(c) $\frac{e}{3}$

(d) $\frac{e}{2}$

(e) $\frac{4}{e}$

15. Using Newton's Method with initial guess $x_1 = 1$, the approximated zero of $f(x) = x^3 - 7$ after two iterations is equal to

(a) $\frac{52}{27}$

(b) $\frac{101}{27}$

(c) $\frac{61}{27}$

(d) $\frac{88}{27}$

(e) $\frac{55}{27}$

16. If $y = \ln\left|\sec\frac{x}{2} + \tan\frac{x}{2}\right|$, then $y'(0) =$

(a) $\frac{1}{2}$

(b) $-\frac{1}{2}$

(c) $-\frac{1}{4}$

(d) 0

(e) $\frac{1}{4}$

17. Which one of the following statements is always true

(a) If f is continuous on $[a, b]$, differentiable on (a, b) and there exists $c \in (a, b)$ such that $f'(c) = 0$, then $f(a) = f(b)$

(b) Newton's Method fails when the initial guess x_1 corresponds to a horizontal tangent line for the graph of f at x_1

(c) If f and g are increasing functions then fg is an increasing function

(d) If the graph of a function has three x-intercepts, then it must have at least two points at which its tangent line is horizontal

(e) The maximum value of $f(x) = x^2$ on the interval $(-3, 3)$ is 9

18. The slope of the tangent line to the curve $4e^{xy} = x^2 - y^2$ at the point $(2, 0)$ is

(a) 1

(b) $\frac{1}{2}$

(c) $\frac{1}{4}$

(d) $\frac{1}{5}$

(e) $\frac{1}{3}$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(a)**	Values at $-1, 0, 1, 2$ are $7, 0, -1, 16$; $M + m = 16 - 1 = 15$.
2	**(c)**	$\frac{y}{x+y} = \frac{1}{4} \Rightarrow x = 3y$, so $\frac{dy}{dt} = \frac{1}{3} \cdot 3 = 1$ m/sec.
3	**(c)**	$y' = -(y/x)^{1/3} = -2$, tangent $y = -2x + 10$, so the intercept is $(5, 0)$.
4	**(e)**	$y' = \frac{2ae^x}{(e^x+a)^2}$, $\frac{2a}{(1+a)^2} = \frac{1}{2} \Rightarrow (a-1)^2 = 0$, $a = 1$.
5	**(a)**	Implicit differentiation at $(-1, 1)$: $-8 + 8y' = -8 + 4y'$, so $y' = 0$.
6	**(b)**	$\frac{dy}{dt} = \frac{2}{3}x^{-1/3}\frac{dx}{dt}$; at $x = 1$, $2 = \frac{2}{3}\frac{dx}{dt}$, so $\frac{dx}{dt} = 3$.
7	**(c)**	MVT on $[0, 1]$: $3 - f(0) \geq 4$, so $f(0) \leq -1$; the largest value is -1.
8	**(e)**	$f' = 2\cos x(1 + 2\sin x)$: $\frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$, sum 5π.
9	**(a)**	$f(0) = \frac{1}{3}$, $\frac{f'}{f}(0) = 6 - \frac{1}{3} = \frac{17}{3}$, so $f'(0) = \frac{17}{9}$.
10	**(a)**	Normal $y = -x + \frac{9}{2}$ meets the parabola again at $(-1, \frac{11}{2})$: $4a + 2b = 7$.
11	**(b)**	$f' = (3x + 1)(x + 1)$ has signs $+, -, +$: local max at -1, local min at $-\frac{1}{3}$.
12	**(e)**	$f(-1) = -5$, $f'(-1) = 6$, so $(f^{-1})'(-5) = \frac{1}{6}$.
13	**(d)**	$y' = \frac{1}{2}\left(\frac{-1}{1-x} - \frac{1}{1+x}\right) + \frac{1}{1+x^2}$, so $y'(0) = -1 + 1 = 0$.
14	**(e)**	$\log_2 c + \frac{1}{\ln 2} = 2 \Rightarrow \ln c = 2\ln 2 - 1$, so $c = \frac{4}{e}$.
15	**(c)**	$x_2 = 3$, $x_3 = 3 - \frac{20}{27} = \frac{61}{27}$.
16	**(a)**	$y' = \frac{\frac{1}{2}\sec\frac{x}{2}\tan\frac{x}{2} + \frac{1}{2}\sec^2\frac{x}{2}}{\sec\frac{x}{2} + \tan\frac{x}{2}}$, so $y'(0) = \frac{1}{2}$.
17	**(b)**	Only the Newton statement is always true: $f'(x_1) = 0$ makes the formula undefined.
18	**(b)**	$4e^{xy}(y + xy') = 2x - 2yy'$ at $(2, 0)$: $8y' = 4$, so $y' = \frac{1}{2}$.